

Salman Haider

GIS Analyst | ArcGIS Pro, Spatial Analysis & Python Automation

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Education



University of Calgary

Calgary, Canada

Master of Geographic Information Systems (MGIS)

08/2024 - 05/2026



Arid Agriculture University

Rawalpindi, Pakistan

Bachelor of Science - Geoinformatics and Earth Observation

09/2016 - 10/2020

Skills

ArcGIS Pro • ArcGIS Online • ArcGIS Enterprise • ArcGIS Field Maps • ArcGIS Dashboards • ArcGIS StoryMaps • ArcGIS Experience Builder • QGIS • Global Mapper • Mango Maps • GeoServer • Leaflet • Mapbox GL JS • Mapbox Studio • MapTiler • Google Earth Engine • ENVI • ERDAS Imagine • Microsoft Office

Python • ArcPy • SQL • JavaScript • HTML/CSS • Flask • GeoPandas • Shapely • GDAL • FME • PostGIS • PostgreSQL • MS SQL • Docker • Tippecanoe

Experience



University of Calgary

Calgary, Canada

GIS Research Assistant

01/2026 - 04/2026

- Analyzed Calgary's public transit access to grocery stores and fresh food sources to identify underserved areas and improve accessibility insights.
- Built a **multimodal transit network** using General Transit Feed Specification (**GTFS**) data, street centreline data, walking connections, and ArcGIS Public Transit Data Model workflows.
- Used ArcGIS Network Analyst to measure 45-minute weekday transit accessibility from residential dissemination areas to grocery stores, supermarkets, and food hubs.
- **Identified 12 dissemination areas with no transit access to fresh food destinations** and 32 with limited access, **affecting more than 33,000 Calgary residents**.



Franchise Ready

Remote

GIS Analyst

11/2021 - 08/2024

- **Delivered 105+ franchise territory mapping projects**, translating client briefs into GIS datasets, territory maps, reports, ArcGIS Dashboards, and MangoMap web map deliverables.
- Used **ArcGIS Pro**, ArcGIS Business Analyst, ABS Census data, demographic enrichment, competitor and POI data, and service-area analysis to build territory layers based on population, income, target demographics, travel distance, and local market conditions.
- Created balanced franchise territories by combining client requirements with demographic patterns, competitor coverage, existing locations, sales information, POI access, and service-area logic.
- Acted as the GIS point of contact during client follow ups, explaining mapping options, data limitations, territory criteria, and recommendations, while automating repeatable GIS tasks with **Python** and **ArcPy** for geocoding, spatial processing, map production, and quality checks.



Greenage Services

Islamabad, Pakistan

GIS Associate

04/2021 - 11/2021

- Created cadastral and land record maps for a government land record digitization project, converting scanned paper maps of state land into structured GIS datasets.
- Digitized state land parcels, boundaries, and cadastral features in **ArcMap**, while creating and maintaining geodatabases with close attention to geometry and attribute accuracy.
- Used **FME Workbench** to convert AutoCAD plans and drawings into land parcel shapefiles, and supported georeferencing of scanned maps, satellite imagery, and drone imagery using **RTK GPS** ground control points.
- Developed **Python** and OpenCV scripts to organize scanned map records and perform **QA/QC** checks on geodatabases and shapefiles for incorrect, missing, or misspelled attribute values before final mapping use.

Experience

MER Solutions

Rawalpindi, Pakistan

GIS Technician

10/2020 - 04/2021

- Worked on the **Benazir Income Support Program** survey boundary mapping project, georeferencing scanned government maps in **ArcMap** and preparing GIS layers for field surveys.
- Created maps showing survey boundaries, roads, landmarks, dwellings, and other key features to support field teams during data collection.
- Linked household survey records with the correct GIS features and checked demographic, economic, land, and asset information for accuracy.
- Reviewed field survey data and produced maps to support data validation, household location verification, land ownership assessment, and final reporting for the government social support program.

Academic Projects

Geo Copilot: Natural Language GIS Tool for ArcGIS Pro

github.com/salman12haider13/arcgis-geo-copilot-llm

ArcGIS Pro | ArcPy | Python | Large Language Models | Geoprocessing | MGIS Capstone

- Developed a custom ArcGIS Pro script tool (Geo Copilot) that enables users to perform geoprocessing tasks using natural language queries, integrating **Large Language Models (GPT-5 and Llama 3.1)** with ArcPy workflows.
- Designed structured reasoning frameworks to translate user intent into executable spatial workflows.
- Evaluated model performance across 36 geospatial test scenarios, achieving up to **91.7%** function selection accuracy and **83.3%** parameter accuracy with GPT-based workflows, and analyzed failure cases related to projection conflicts and multi-step reasoning.

Ord's Kangaroo Rat Habitat Suitability Model

ArcGIS Pro | Suitability Modelling | Raster Analysis | Aspect Analysis | Habitat Mapping | MGIS Project

- Built an ArcGIS Pro habitat suitability model using **Model Builder** for Ord's Kangaroo Rat using DEM, active sand dunes, ecological range sites, unpaved roads, fireguards, and known species occurrence data from CFB Suffield.
- Prepared the analysis datasets by clipping, projecting, and standardizing DEM, road, fireguard, sand dune, and species occurrence layers to a common study area.
- Created the primary habitat model by identifying areas near active sand dunes, extracting dune tops with DEM focal statistics, and selecting south to southwest-facing dune aspects linked to suitable habitat.
- Developed the secondary habitat model using sandy road and fireguard conditions, dune proximity, and road orientation, then combined final rasters using a 70 percent primary habitat and 30 percent secondary habitat weighting.

Brazeau Dam Land Cover Classification and NDVI Time-Series Analysis

Google Earth Engine | JavaScript | ArcGIS Pro | Landsat 5, 7 and 8 | Image Classification | NDVI | MGIS Project

- Built a 4,567 square kilometre remote sensing study area around Brazeau Dam in Google Earth Engine using Landsat 5, Landsat 7, and Landsat 8 imagery.
- Created unsupervised and supervised land cover classifications using **K-Means clustering**, **CART classification**, training samples, validation samples, class remapping, and confusion matrix accuracy assessment.
- Produced NDVI median composites for 1990, 2005, and 2020 to compare vegetation conditions across a long-term time series.
- **Achieved 71.875 percent overall classification accuracy** and found vegetation decline by 2005 followed by recovery trends by 2020.

Academic Projects

Namal Lake Desiccation Assessment, 2000 to 2024

Remote Sensing | Google Earth Engine | Python | ArcGIS Pro | ERA5-Land | MNDWI | NDVI | OLS Regression | MGIS Project

- Processed Landsat 7, Landsat 8, and Sentinel-2 imagery in Google Earth Engine to measure Namal Lake surface area change from 2000 to 2024.
- Used **MNDWI** for annual water extraction and NDVI to estimate agricultural expansion around the lake, then built annual median composites and environmental time-series layers.
- Integrated ERA5-Land climate variables including temperature, precipitation, watershed evaporation, and lake surface evaporation, then tested relationships using **standardized OLS regression** in Python.
- Found a statistically significant decline in lake surface area, with agricultural expansion emerging as the strongest negative predictor and an **adjusted R² of 0.594**.

LiDAR and Object-Based Conifer Tree Height Classification

Remote Sensing | ENVI 6.0 | ArcGIS Pro | LiDAR CHM | QuickBird Imagery | Object-Based Image Analysis | SVM | KNN | MGIS Project

- Classified land cover and coniferous tree heights in a Campbell River, British Columbia study area using ENVI 6.0, ArcGIS Pro, QuickBird imagery, and a 1 metre LiDAR Canopy Height Model.
- Used NIR, RGB, and NDVI bands for example-based feature extraction, image segmentation, and training sample creation for roads, large trees, small trees, and clear-cut areas.
- Compared **KNN** and **SVM** classification methods and selected SVM because it performed better for identifying smaller clear-cut patches.
- Created nine LiDAR-based tree height classes from 0 to 45 metres, finding that the 25 to 30 metre class covered the largest area at 307,565 square metres.

Logistics Routing MVP

github.com/salman12haider13/Logistics-Routing-MVP-Demo

PostGIS | pgRouting | PostgreSQL | Flask | Docker | Leaflet | OpenStreetMap | GeoJSON | MGIS Project

- Built a self-hosted web GIS routing prototype using OpenStreetMap road data, PostgreSQL, PostGIS, pgRouting, Flask, Docker, and Leaflet.
- Designed the full routing workflow from raw OSM PBF data to a routable PostGIS network, including road filtering, topology creation, source and target nodes, travel-time costs, one-way handling, and spatial indexing.
- Developed a Flask backend API that accepts start and end points, routing mode, blocked polygons, traffic-delay points, and route alternative settings.
- Created an interactive Leaflet routing console that returns GeoJSON route geometry, route summaries, distance, travel time, route steps, blocked-zone routing behaviour, and alternative route options.

GEOSS Remote Sensing MVP Workflow for Vegetation Risk and Property Maintenance

Remote Sensing | Python | QGIS | PostgreSQL | PostGIS | Mapbox | AWS | Segment Anything Model | MGIS Project

- Designed a remote sensing MVP workflow for aerial vegetation identification, property risk assessment, and property maintenance planning using a sampled Vancouver study area.
- Proposed a processing pipeline using high-resolution orthophotos, Microsoft building footprints, PostGIS storage, Mapbox visualization, Python, QGIS, AWS processing, and SAM-based segmentation.
- Designed separate workflows for building segmentation, tree canopy extraction, canopy proximity analysis, and moss detection using buffers, clipped imagery, NDVI logic, and GeoJSON outputs.
- Framed the workflow around practical use cases including tree failure risk, moss growth, property damage, insurance risk, and municipal planning support, without presenting it as a full production deployment.

Courses

FME Desktop: Data Translation for AEC

LinkedIn

Creating a Map Layout

Esri

Labeling Map Features

Esri

Creating a Web Map in ArcGIS Pro

Esri

Terrain Analysis Using ArcGIS Pro

Esri

Learning ArcGIS Python Scripting

LinkedIn

Courses

Telling Stories with GIS Maps

Esri

Using Python for Automation

LinkedIn

Advanced Python: Working with
Databases

LinkedIn

Generating Service Areas Using ArcGIS
Pro

Esri

Suitability Modeling: Creating a
Weighted Suitability Model

Esri

Performing Supervised Object-Based
Image Classification

Esri

PostgreSQL: Advanced Queries

LinkedIn

Location-Enabling Data

Esri

Building Geoprocessing Models Using
ArcGIS Pro

Esri

Suitability Modeling: Creating a Simple
Suitability Model

Esri

Performing Accuracy Assessment for
Image Classification

Esri

Performing Supervised Pixel-Based
Image Classification

Esri

Editing Basics in ArcGIS Pro

Esri

Mapping Addresses and Places

Esri

Creating Optimized Routes Using
ArcGIS Pro

Esri

Getting Started with Mapping and
Visualization

Esri

Finding the Closest Facilities Using
ArcGIS Pro

Esri

Coordinate Reference Systems,
Projections, and Spatial Data
Representation

GEO University